

ACP

AARAMBH CHAPTERWISE PROBLEMS

Class-9th

DO NOT CROSS POLICE LINE

DO NOT CROSS POLICE LINE

The Fundamental Unit of Life

PRASHANT KIRAD



The fundamental unit of life

 (Options acche se padhna)

MULTIPLE CHOICE QUESTIONS (1 Mark)

1. Which of the following is the basic unit of life?
 - A. Tissue
 - B. Organ
 - C. Cell
 - D. Organ System

2. Which cell organelle contains digestive enzymes and helps in the digestion of cellular waste?
 - A. Golgi apparatus
 - B. Endoplasmic reticulum
 - C. Mitochondria
 - D. Lysosomes

3. The shrinking of the cytoplasm away from the cell wall due to water loss is known as:
 - A. Plasmolysis
 - B. Endocytosis
 - C. Exocytosis
 - D. Turgidity

4. Who discovered the cell?
 - A. Robert Brown
 - B. Antonie van Leeuwenhoek
 - C. Robert Hooke
 - D. Rudolf Virchow

5. Which organelle is known as the 'powerhouse of the cell'?
 - A. Golgi apparatus
 - B. Lysosome
 - C. Endoplasmic reticulum
 - D. Mitochondria

6. Which of these is selectively permeable?

- A. Cell wall**
- B. Plasma membrane**
- C. Nucleus**
- D. Cytoplasm**

7. Which process involves movement of water across a semi-permeable membrane?

- A. Diffusion**
- B. Osmosis**
- C. Active transport**
- D. Plasmolysis**

8. A plant cell placed in a hypertonic solution will:

- A. Swell**
- B. Burst**
- C. Undergo plasmolysis**
- D. Remain the same**

9. In plant cells, which organelle stores starch, oils, and proteins?

- A. Chloroplast**
- B. Chromoplast**
- C. Leucoplast**
- D. Lysosome**

10. Chromosomes are made of:

- A. DNA**
- B. RNA**
- C. DNA and proteins**
- D. Proteins only**

11. Which organelle helps in packaging and secretion of substances?

- A. Mitochondria**
- B. Ribosomes**
- C. Golgi apparatus**
- D. Plastids**

12. What is an example of a prokaryotic cell?

- A. Protozoa**
- B. Fungi**
- C. Plant cell**
- D. Bacteria**

13. What is the function of the cytoplasm in a cell?

- A) Storage of genetic material
- B) Site of photosynthesis
- C) Location of cellular metabolism and organelle support
- D) Regulation of cell functions

14. The organelle responsible for synthesis of proteins is:

- A. Ribosome
- B. Lysosome
- C. Golgi body
- D. Vacuole

 (ho jaayenge aaram se)

VERY SHORT ANSWER TYPE QUESTIONS (1 Marks)

15. What is osmosis?

16. What do lysosomes do ?

17. What is plasmolysis?

18. Name the semi-permeable membrane of the cell.

19. Which cell organelle has its own DNA?

20. Which cell organelle is absent in animal cells but present in plant cells?

21. Who discovered the cell?

22. What is the basic unit of life?

23. Name the jelly-like substance present inside the cell.

24. What controls the activities of a cell?

25. Which cell organelle is known as the powerhouse of the cell?

26. What is the function of the cell wall?

27. Name the organelle responsible for photosynthesis.

 (pahle points socho firr likho)

SHORT ANSWER TYPE QUESTIONS (2 - 3 Marks)

28. What are the differences between prokaryotic and eukaryotic cells?
(Mention at least three differences.)

29. Explain the structure and function of the plasma membrane.

30. What are the functions of the nucleus in a cell?

31. Explain the process of osmosis with an example.

32. Why are mitochondria known as the powerhouse of the cell?

33. Write three differences between plant and animal cells.

34. What happens to a plant cell when it is placed in a:

- a) hypertonic solution
- b) hypotonic solution
- c) isotonic solution

35. What is the significance of mitosis?

 (pahle points socho firr likho)

Long Answer Type Questions (5 Marks)

36. i) Justify the statement “a cell is the structural and functional unit of life”.

ii). Explain the structure and functions of different cell organelles.

37. Explain the structure and function of the plasma membrane. Why is it called a selectively permeable membrane?
38. Explain the structure and function of the nucleus. Why is it called the control center of the cell?
39. Write a detailed note on the Endoplasmic Reticulum. Differentiate between Rough ER and Smooth ER.
40. What are plastids? Describe the structure and functions of different types of plastids.
41. Describe the structure of the plasma membrane. How does it help in the transport of substances across the cell?
42. Explain the process of osmosis and diffusion. Give suitable examples.
43. Explain the terms plasmolysis, endosmosis, and exosmosis with examples.
44. Describe the functions of the following cell organelles: mitochondria, Golgi apparatus, lysosomes, vacuoles, ribosomes.



Case Study/Source Based Question

45. Case Study 1:

Case:

Riya placed a few raisins in plain water and left them overnight. The next day, she observed that the raisins had swollen. Curious, she placed the swollen raisins in a concentrated sugar solution, and after some time, they shrank again.

Questions:

- (a) What process caused the raisins to swell?
- (b) Why did the raisins shrink in the sugar solution?
- (c) Define osmosis and explain its role in the experiment.
- (d) What would happen if dry raisins were placed in saltwater?

46. Case Study 2:

Case:

A scientist is observing onion peel cells under a microscope. She notes that the cells have a well-defined cell wall, a large central vacuole, and green-colored structures in some cells when exposed to light.

Questions:

- (a) Which cell organelle is responsible for the green color observed in some cells?
- (b) What is the function of the large central vacuole in plant cells?
- (c) Identify one structural difference between plant and animal cells visible in this observation.
- (d) Why are onion cells considered eukaryotic?

 (yaha Marks katate h)

Assertion And Reasons Questions

47. Assertion (A): Mitochondria are known as the powerhouse of the cell.

Reason (R): Mitochondria help in the synthesis of proteins

48. Assertion (A): Plant cells do not burst when placed in hypotonic solutions.

Reason (R): The presence of a rigid cell wall prevents the cell from bursting.

49. Assertion (A): Ribosomes are involved in protein synthesis.

Reason (R): Ribosomes are found only in animal cells.

50. Assertion (A): When a plant cell is placed in a hypertonic solution, it undergoes plasmolysis.

Reason (R): The cell wall of plant cells dissolves in salt solutions.

51. Assertion (A): Lysosomes are also called “suicidal bags” of the cell.

Reason (R): Lysosomes contain enzymes that can digest the entire cell if they burst.

52. Assertion (A): The nucleus controls all cellular activities.

Reason (R): It contains ribosomes that synthesize proteins for the cell.

53. Assertion (A): Plasma membrane is semi-permeable.

Reason (R): It allows only water molecules to enter and exit the cell.



PK Special [KBC]

54. A scientist discovered a new cell that lacked a nucleus but had DNA. Suggest which type of organism it could belong to and justify.
55. A plant cell is placed in a concentrated sugar solution. Describe the series of events that will occur inside the cell and relate it to plasmolysis.
56. Why is it dangerous to store pickles or curd in metal containers?
57. If a cell loses its mitochondria but retains all other organelles, which life processes will be directly affected and why?
58. Compare the role of the endoplasmic reticulum with a manufacturing and delivery system in a factory. Which type (smooth or rough) would be more critical for a liver cell and why?
59. A cell continues to swell even though it is in a solution that contains salt. What could be the possible reason?
60. Predict the consequences if the lysosomes of a white blood cell suddenly burst inside the cell.
61. How would a cell change if its Golgi apparatus was removed?
62. Suppose a cell lacks ribosomes. What impact would that have on the nucleus and overall functioning of the cell?
63. Why do onion peel cells not burst when kept in water, while animal cells can?
64. How is the structural difference between plant and animal cells responsible for the way each responds to a hypotonic solution?
65. If you had to design a cell resistant to acidic environments, what changes would you make in its organelles and why?

SOLUTIONS

MULTIPLE CHOISE TYPE QUESTIONS

1. C
2. B
3. B
4. C
5. D
6. B
7. B
8. C
9. C
10. C
11. C
12. D
13. C
14. A

VERY SHORT ANSWER TYPE QUESTIONS

15. Osmosis is the movement of water molecules from a region of higher water concentration to a region of lower water concentration through a semi-permeable membrane.

16. Lysosomes help in digestion and removal of waste; they break down damaged cell parts and foreign materials. They are also called the "suicide bags" of the cell.

17. Plasmolysis is the process in which the cell membrane pulls away from the cell wall due to loss of water when a plant cell is placed in a hypertonic solution.

18. The plasma membrane (or cell membrane) is the semi-permeable membrane of the cell.

19. The mitochondria and chloroplasts are cell organelles that have their own DNA.

20. Chloroplast is present in plant cells but absent in animal cells.

SOLUTIONS

17. Plasmolysis is the process in which the cell membrane pulls away from the cell wall due to loss of water when a plant cell is placed in a hypertonic solution.

18. The plasma membrane (or cell membrane) is the semi-permeable membrane of the cell.

19. The mitochondria and chloroplasts are cell organelles that have their own DNA.

20. Chloroplast is present in plant cells but absent in animal cells.

21. Robert Hooke

22. Cell

23. Cytoplasm

24. Nucleus

25. Mitochondria

26. The cell wall provides structural support and protection to the plant cell. It also helps maintain the shape of the cell.

27. Chloroplast

SHORT ANSWER TYPE QUESTIONS:

28.	Prokaryotic Cells	Eukaryotic Cells
	No true nucleus (nucleoid present)	True nucleus with nuclear membrane
	Membrane-bound organelles are absent	Membrane-bound organelles are present
	Generally smaller and simpler	Generally larger and more complex

29. The plasma membrane is a thin, flexible, semi-permeable membrane made of a phospholipid bilayer with embedded proteins.

Function: It controls the movement of substances in and out of the cell and helps maintain the internal environment of the cell.

SOLUTIONS

30. (i) Controls all cellular activities (like a control center)

(ii) Contains genetic material (DNA)

(ii) Helps in cell division and inheritance

31. Osmosis is the movement of water molecules through a semi-permeable membrane from a region of high water concentration to a region of low water concentration.

Example: When raisins are placed in water, water enters the raisins by osmosis and they swell up.

32. Mitochondria generate energy in the form of ATP (Adenosine Triphosphate) during cellular respiration, which is used for all cellular activities.

33.	Plant Cells	Animal Cells
	Have a cell wall	No cell wall
	Contain chloroplasts for photosynthesis	Do not contain chloroplasts
	Large central vacuole	Small or no vacuoles

34. a) Hypertonic solution – Cell loses water and undergoes plasmolysis.

b) Hypotonic solution – Cell gains water and swells (may become turgid).

c) Isotonic solution – No change; water movement is balanced.

35. Mitosis is important because:

- It helps in the growth of an organism.
- It replaces old, dead, or damaged cells.
- It maintains the chromosome number in daughter cells (genetic stability).

LONG ANSWER TYPE QUESTIONS:

36. i) All living organisms are made of cells. They perform all vital functions (like respiration, digestion, excretion, etc.). All life processes occur within cells, making them the basic structural and functional units of life.

ii). Nucleus – Controls activities, contains DNA

- Mitochondria – Produces energy (ATP)
- Golgi apparatus – Packages and transports materials
- Lysosomes – Digests waste and old organelles
- Ribosomes – Synthesizes proteins
- Endoplasmic Reticulum (ER) – Transport system of the cell

SOLUTIONS

- Plastids (in plants) – Chloroplasts do photosynthesis
- Vacuoles – Storage of substances and maintaining cell shape

37. Structure of Plasma Membrane:

The plasma membrane is the outermost covering of the cell. It is made up of proteins and lipids and has a flexible, thin, and delicate structure. It is also known as the cell membrane.

Function of Plasma Membrane:

- It protects the contents of the cell.
- It provides shape and support to the cell.
- It regulates the entry and exit of substances in and out of the cell.
- It maintains the internal environment of the cell.

Why it is called selectively permeable:

The plasma membrane is called selectively permeable because it allows only certain substances (like oxygen, water, and nutrients) to pass through it, while restricting others. This selective movement helps the cell maintain balance and function properly.

38. The nucleus is a double-membrane organelle containing nucleoplasm, chromatin (DNA), and a nucleolus.

It regulates all cell functions, stores genetic material, and controls cell growth, metabolism, and reproduction. It's called the control center because it directs all cellular activities.

39. The Endoplasmic Reticulum (ER) is a network of membranes that helps in the transport of substances inside the cell.

- Rough ER (RER): Has ribosomes; synthesizes proteins.
- Smooth ER (SER): No ribosomes; synthesizes lipids, detoxifies chemicals.

Functions: Protein and lipid synthesis, intracellular transport.

40. Plastids are double-membraned organelles found in plant cells. They contain pigments and play a role in storage and synthesis.

Types:

- Chloroplasts – Contain chlorophyll, perform photosynthesis.
- Chromoplasts – Contain colored pigments (yellow, red); give color to fruits and flowers.
- Leucoplasts – Colorless; store starch, oils, proteins.

SOLUTIONS

41. Structure of Plasma Membrane: The plasma membrane is composed of a phospholipid bilayer, where the hydrophobic (water-repellent) tails of phospholipids face inward, and the hydrophilic (water-attracting) heads face outward. This structure forms a semi-permeable barrier. Embedded in this bilayer are proteins, which may be integral (spanning across the membrane) or peripheral (on the surface). The membrane also contains cholesterol molecules that help in maintaining its fluidity.

Functions in Transport:

- The plasma membrane plays a crucial role in the transport of substances:

Passive Transport (e.g., Diffusion, Osmosis):

- In diffusion, small molecules like oxygen and carbon dioxide pass through the membrane without energy expenditure.
- Osmosis is the movement of water molecules from a region of lower solute concentration to higher solute concentration through the membrane.

Active Transport:

- Active transport requires energy (ATP) to move substances against the concentration gradient. For example, the sodium-potassium pump actively moves sodium ions out of the cell and potassium ions into the cell.

Endocytosis and Exocytosis:

- Large molecules, such as food or waste products, are transported into and out of the cell through processes like endocytosis (cell engulfing substances) and exocytosis (substances leaving the cell in vesicles).
- Thus, the structure of the plasma membrane ensures selective permeability, allowing only certain molecules to pass while keeping others out, thus maintaining the cell's internal environment and allowing necessary substances to move in and out.

42. Diffusion: Movement of particles from high to low concentration.

- Example: Perfume scent spreading in air.

Osmosis: Movement of water through a semi-permeable membrane from high to low water concentration.

- Example: Swelling of raisins in water.

43. Plasmolysis: Cell membrane shrinks away from the wall due to water loss in hypertonic solution. E.g., plant cell in salt water.

Endosmosis: Water enters the cell in hypotonic solution. E.g., raisin in plain water

SOLUTIONS

swells.

- **Exosmosis:** Water leaves the cell in hypertonic solution. E.g., swollen raisin placed in salt water shrinks.

44. Describe the functions of the following cell organelles:

- **Mitochondria:** Generate energy (ATP) via respiration.
- **Golgi apparatus:** Processes, packages, and ships proteins/lipids.
- **Lysosomes:** Break down waste and damaged organelles.
- **Vacuoles:** Store food, water, and waste; maintain pressure.
- **Ribosomes:** Build proteins by linking amino acids.

CASE STUDY/SOURCE BASED QUESTION:

45. Case Study 1:

(a) The process that caused the raisins to swell is osmosis. Water entered the raisins from the surrounding water through their semi-permeable membrane.

(b) The raisins shrank in the sugar solution because the solution was hypertonic (higher concentration of solute). Water moved out of the raisins into the sugar solution, causing them to shrink.

(c) Osmosis is the movement of water molecules from an area of higher concentration to an area of lower concentration through a semi-permeable membrane. In this experiment, osmosis caused water to enter the raisins when placed in plain water (hypotonic solution), making them swell, and water to exit the raisins when placed in a sugar solution (hypertonic solution), causing them to shrink.

(d) If dry raisins were placed in saltwater (a hypertonic solution), water would leave the raisins, causing them to shrink or undergo plasmolysis.

46. Case Study 2:

(a) The organelle responsible for the green color in some cells is the chloroplast, which contains the green pigment chlorophyll.

(b) The large central vacuole in plant cells serves to store water, nutrients, and waste products. It also helps maintain turgor pressure, which provides structural support to the cell.

SOLUTIONS

(c) A key structural difference observed is the presence of a cell wall in the onion cells, which is absent in animal cells.

(d) Onion cells are considered eukaryotic because they have a true nucleus and membrane-bound organelles, which are characteristics of eukaryotic cells.

ASSERTION AND REASONS TYPE QUESTIONS:

47. A

48. A

49. C

50. C

51. A

52. C

53. C

SOLUTIONS

PK Special [KBC]

54. It likely belongs to a prokaryotic organism (e.g., bacteria). Prokaryotes lack a membrane-bound nucleus, and their DNA is found freely in the cytoplasm. This is a key feature distinguishing them from eukaryotes.

55. Water moves out of the cell through osmosis (since the external solution is hypertonic). The cytoplasm shrinks, and the plasma membrane pulls away from the cell wall. This is called plasmolysis.

56. Pickles and curd are acidic. Acids react with metals (like iron or aluminum), forming toxic salts, which can be harmful if consumed.

57. Cellular respiration will be affected because mitochondria produce ATP (energy). Without them, the cell cannot generate sufficient energy to perform activities like active transport or growth.

58. The ER acts like a factory system — Rough ER manufactures proteins (with ribosomes), and Smooth ER helps in lipid synthesis and detoxification. For liver cells, Smooth ER is more critical because it detoxifies harmful substances and helps in lipid metabolism.

59. The salt concentration in the solution is still less than the concentration inside the cell (hypotonic solution). Hence, water continues to enter the cell by osmosis, causing it to swell.

60. Lysosomes contain digestive enzymes. If they burst, these enzymes would digest the cell's internal components, causing self-destruction. This is why lysosomes are called “suicidal bags” of the cell.

61. The cell would be unable to package and transport proteins and lipids. Secretion of enzymes and hormones would stop, affecting cellular function and communication.

62. Ribosomes are responsible for protein synthesis. Without them, no proteins would be made, so cell growth, repair, and enzyme production would halt. The nucleus would still store DNA but couldn't express it effectively.

SOLUTIONS

63. Onion cells have a rigid cell wall that prevents bursting. Animal cells lack a cell wall, so when excess water enters, they can swell and burst.

64. Plant cells have a cell wall that limits swelling and prevents bursting in a hypotonic solution. Animal cells lack this wall and are more likely to burst as water enters unchecked.

65. Strengthen the plasma membrane with extra lipids to resist acid entry.

- **Increase the number of lysosomes to digest harmful substances.**
- **Add a thicker or chemically-resistant cell wall (like in some bacteria).**
- **Include protective enzymes that neutralize acids (e.g., acidophilic adaptations).**